

Ludic Ambiguity on the Wrist: How an “Imperfect” Stress Avatar Becomes a Social Play Mechanic

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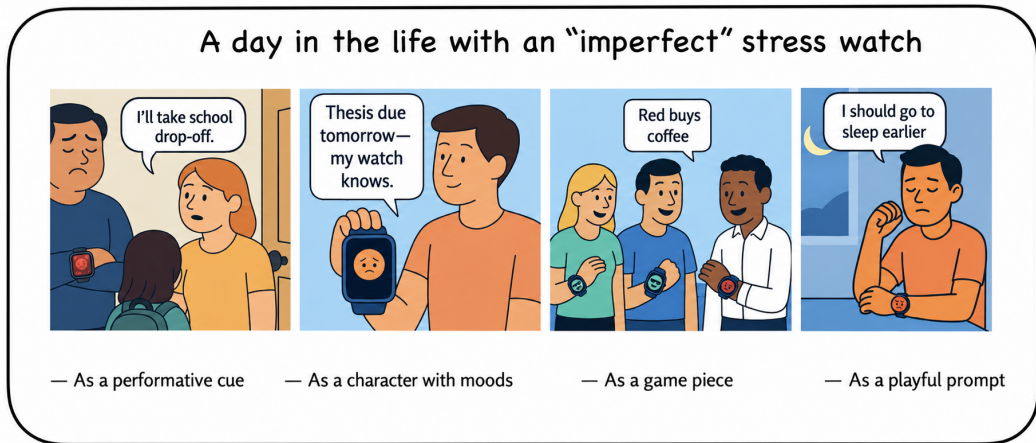


Fig. 1. Four ways users appropriate an “imperfect” stress watch: as a performative cue, a companion, a game piece, and a reflective prompt. These roles make a contestable biosignal socially usable while raising risks around visibility, uncertainty, and emotional burden.

Wearable biosignals increasingly circulate as social cues, not just personal measurements. We examine StressWatch, a third-party smartwatch app that estimates stress from heart-rate-derived signals and renders it as a glanceable animated avatar on the watch face. Using a multi-method qualitative study—238 App Store reviews, 174 public posts on RED (XiaoHongShu), and 18 semi-structured interviews across North America, China, and Europe—we examine how users navigate the gap between what the system infers and what they feel. We name this gap a *playable gap*: a space of interpretive openness that becomes a resource for social play. The playable gap supports five recurring forms of everyday play: performative (staging and signaling states), narrative (jokes, folk theories, and collective sensemaking), competitive/cooperative (challenges, teasing, and solidarity rituals), companionship (personification and care), and reflective (micro-experiments and pattern seeking). We argue that, in this case, such biosignal displays function as *ambiguity-sustained ludic systems*, while also introducing risks around visibility, uncertainty burden, and social pressure. We conclude with design implications for legible uncertainty and consensual sharing.

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CCS Concepts: • **Human-centered computing** → **Interaction design theory, concepts and paradigms**; *Empirical studies in HCI*; *Ubiquitous and mobile computing*; *Social computing*; Human-computer interaction (HCI).

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1 Introduction

Wearables have become everyday companions, and increasingly everyday playthings. People turn step counts into friendly rivalries, treat notifications as prompts for mini-challenges, and circulate biosignal screenshots as jokes, check-ins, and memes [6, 13, 28]. These practices show that health technologies are not only instruments for self-tracking, but also materials for *play*: users improvise rules, stage performances, and make meaning together through lightweight, situated interaction. Yet much research on wearables still foregrounds validation of inferred states and whether people act on feedback as intended [12, 22, 36]. We argue that ambiguity in a wearable stress display can function as a resource for playful, social interaction: when a biosignal cue remains open to interpretation, it creates a *playable gap* through which people joke, compare, test, and negotiate meaning together. Rather than directly representing an internal state, the biosignal here works as a socially interpretable cue that invites interaction. In this case, the signal becomes a socially usable play resource rather than only a measurement to verify.

Stress tracking is a particularly revealing site for this question. Stress is difficult both to define and to sense, and mismatches between what a device infers and what people feel are common [8, 32]. We use three terms with distinct meanings throughout the paper. *Ambiguity* refers to interpretive openness: the display affords multiple plausible meanings. *Inaccuracy* refers to a perceived mismatch between system output and lived experience. *Latency* refers to delay between experience and system feedback. Our focus is not error in general, but ambiguity as the key condition that makes a stress cue contestable and socially usable. In many stress-tracking designs, uncertainty is treated as a problem to be minimized because it can undermine trust and drive disengagement. We instead ask what people do when a stress cue is easy to show to others yet remains open to interpretation. In such cases, visible uncertainty becomes an everyday “puzzle”—a cue people interpret, test, and negotiate with others—sometimes playfully, sometimes tensely.

This possibility is especially salient when stress is rendered through expressive characters rather than numeric dashboards. Within avatar-based representations, feedback can function less like a report and more like a micro-game state: a glanceable cue that invites interpretation, teasing, and low-stakes experimentation. A watch face can become a stage, an avatar a companion, and an ambiguous cue a mechanic mobilized into rituals, challenges, and shared stories. Prior HCI and game design work argues that ambiguity can be an intentional design resource, inviting interpretation, reflection, and playful meaning-making rather than simple clarity [2, 14]. From a game design perspective, meaningful play emerges when participants interpret feedback and construct meaning through interaction, even when system behavior is not fully deterministic [37]. Together, these perspectives position interpretive openness, in this case, as the condition that turns biosignal feedback into social interaction rather than mere display.

We study these dynamics through *StressWatch* [33], a third-party smartwatch app that estimates stress from heart-rate-derived signals and renders the result as a cartoon avatar embedded on the watch face (Figure 2). The avatar shifts color and expression (e.g., green/calm, orange/elevated,

red/high), offering a glanceable character that users can reveal with a wrist turn and circulate via screenshots. While the app is framed as a more playful alternative to clinical dashboards [13, 15], users in our dataset took that framing further: they treated the avatar as a game piece for low-stakes challenges, a performative “mood badge” for social signaling, a pet-like companion to soothe, and a meme-able character whose apparent “glitches” sparked folk theories and collective worldbuilding. In this case, these practices are not peripheral to use; they show how users appropriate uncertainty itself as a shared interaction resource.

Prior work has examined gamification and social support in health technologies [5, 22], playful biofeedback and shared cues [13], and domestic play formats that gather households around lightweight co-present interaction [20, 24]. However, a gap remains in understanding how people *spontaneously* transform ambiguous affective health feedback into sustained play practices with recognizable play loops (showing, comparing, teasing, caring), rather than only sensemaking or compliance. This gap matters for CHI PLAY because, in cases like this, biosignal wearables become socially visible artifacts whose meanings are produced through interaction with others, not solely through individual reflection.

We report a multi-method qualitative study drawing on App Store reviews, public posts on RED (XiaoHongShu), and 18 interviews. Across sources, the stress avatar is repeatedly mobilized as a social play artifact: people perform states by raising their wrist, trade screenshots as story fragments, invent cooperative and competitive rituals, and personify the character as a companion that can be “taken care of.” These practices persist even when readings remain contestable, showing that participants appropriate uncertainty as a shared interaction resource rather than treating it only as a defect to resolve.

Our central argument is that, in this instrumental case, the avatar functions as an *ambiguity-sustained ludic system*: a biosignal display whose social use is organized through the interpretive openness of its readings (Figure 1). The mechanism we identify is the playable gap between system output and lived experience. This gap is made legible and actionable through four design mechanics—visibility, contestability, circulation, and persistence. We see it enacted in five empirical play forms: performative, narrative, competitive/cooperative, companionship, and reflective. This contributes to CHI PLAY theorizing by showing how physiological feedback—typically framed as measurement—can become a playable resource that supports interpretive, social, and emergent forms of play beyond conventional games and beyond precision-centric health tracking. At the same time, this reframing raises design and ethical questions: when does playful stress signaling support coping and connection, and when does it create anxiety, social pressure, or unwanted visibility? Ambiguity can help sustain play when feedback is socially legible, easily shared, and open to negotiation.

In this paper, we identify play when users enact repeatable, socially negotiated routines around the avatar (e.g., show–react–narrate–compare–care/test) that are framed as low-stakes and voluntary by participants. Across our findings, play takes the form of (i) performative showing and staging of states, (ii) narrative sharing and collective storytelling, (iii) cooperative and competitive challenges, (iv) companionship and personification of the avatar, and (v) reflective inquiry through small experiments. These forms specify how ambiguity becomes actionable as play in this particular system. We address:

- **RQ1:** How do people interpret, share, and use an avatar-based stress cue in everyday life when it remains contestable?
- **RQ2:** What mechanics and boundary conditions make these routines supportive vs harmful?

We acknowledge that *StressWatch* is designed with a playful aesthetic. Our contribution is not that play emerged simply because the interface looked playful; rather, in this case, ambiguity

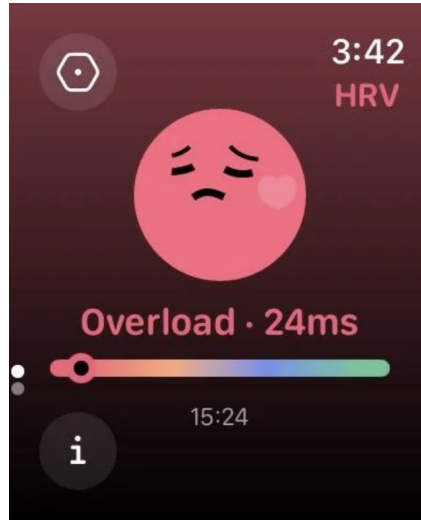


Fig. 2. The *StressWatch* avatar on the smartwatch face during an inferred high-stress state. The character’s color (green/calm, orange/elevated, red/high) and facial expression (smile, neutral, frown) together communicate a qualitative stress state at a glance. A secondary HRV-derived readout and intensity scale are also shown, but the primary cue is the expressive avatar rather than a numeric stress score. Because the avatar occupies the primary watch face, it is surfaced through a routine wrist turn and is easily shareable via screenshots—making it socially showable in co-present settings (raising a wrist to reveal a state to a partner or colleague) and circulatable at a distance (sending a snapshot to a group chat). These representational choices—a qualitative, character-based cue rather than a precise numeric verdict, an expressive character rather than a dashboard, and a watch-face placement rather than a hidden menu—are central to the play practices we report.

shaped which play forms emerged, how they were sustained when the signal felt wrong, and where they broke down. Our data point in this direction: many of the most circulated moments in our corpus—the “dramatic Tamagotchi” nicknames, the karaoke and gaming memes, the folk theories about “it only sees stillness”—turn explicitly on the *mismatch* between reading and felt experience, not the cute character alone. This suggests that a more accurate system with the same avatar design may produce different practices, because the contestability of the reading appears to be doing work that the playful aesthetic by itself does not.

Our contributions are threefold. **Conceptual:** Develop the concept of *ambiguity-sustained ludic systems* through an instrumental case of an avatar-based stress wearable, and identify the playable gap as the mechanism through which users appropriate uncertainty itself (not just the data) as a play resource. **Empirical:** Identify five recurring social play forms around an affective avatar (performative, narrative, competitive/cooperative, companionship, reflective) supported across reviews, RED, and interviews. **Design:** Provide play-mechanic and safeguard pairings (e.g., visibility/audience control; persistence/anti-spiral off-ramps; contestability/explain-on-demand) to support consensual, low-stakes play.

2 Related Work

This paper brings together work on play and ambiguity, wearable interaction and expressive displays, lived and interpersonal informatics, and stress inference and its risks. We first situate play as an everyday interpretive practice rather than a property of formal games. We then consider

wrist-worn wearables as socially expressive media. Third, we draw on lived and interpersonal informatics to understand appropriation and social sensemaking around personal data. Finally, we review stress inference as an inherently uncertain biosignal domain and outline the risks associated with ambiguous affective feedback.

2.1 Play Beyond Games and Ambiguity as Resource

Games and play scholarship has emphasized that meaning emerges through interpretation: people make sense of feedback, characters, and cues in context rather than through mechanical precision alone [19, 37]. Recent CHI PLAY work extends these perspectives to non-traditional contexts, examining how playful mechanics can support reflection, emotional regulation, and everyday interaction beyond entertainment-focused games [18]. Xu et al. show that embodied play can be organized through deliberately designed movement logics, focusing on how structured bodily mechanics shape gameplay in AR systems [40]. In contrast, our findings show play emerging less from prescribed movement mechanics than from interpretive openness and socially negotiated readings of a biosignal cue. Ambiguity has similarly been framed as a design resource that invites interpretation and encourages people to construct meaning rather than passively receive information [2, 14]. Recent work on self-tracking wearables makes this especially relevant here: Di Lodovico et al. show that ambiguity can be designed into wearable tracking experiences to support interpretation rather than false precision [7]. However, that work tends to focus on ambiguity as a design property of self-tracking interfaces, whereas we show how ambiguity becomes a playable social mechanism through a space that people act on together. CHI PLAY scholarship further shows how participants negotiate system behavior through cultural expectations and prior experiences when designs introduce uncertainty or deceptive cues [16]. While ambiguity has been widely discussed as a resource and play as an interpretive practice, we still lack empirical accounts of how ambiguous biosignal feedback becomes sustained everyday play.

2.2 Wearables as Playful and Social Media

Wearables enable embodied, glanceable interaction: they are always-on, easy to incorporate into gesture, and readily surfaced in co-present moments through a simple wrist turn [1, 15]. Frey et al.'s *Breeze* shows that biofeedback can operate as social communication, allowing close ties to share and feel one another's bodily states through wearable feedback [13]. In contrast, *Breeze* largely treats biosignals as a channel for interpersonal awareness, whereas our findings show how ambiguity in an expressive stress avatar creates a shared interpretive space for teasing, comparison, and low-stakes rituals. Liang et al. show that smartwatch health games benefit from carefully timed interactions aligned with routines, focusing on how structured timing mechanics shape engagement [27]. In contrast, our findings show socially negotiated play emerging from interpretive flexibility rather than from optimizing when the system intervenes. Similarly, recent CHI PLAY work such as *Ghost Gait* shows how wearable fitness feedback can become memetic and accountability-oriented; in contrast, our case focuses on how ambiguity in biosignal feedback becomes a resource for socially negotiated play [39]. Recent CHI PLAY work shows how domestic and co-present play can be scaffolded through lightweight infrastructures, such as movement prompts framed as playful micro-rituals [20], tangible story-based interactions that invite shared narration [3], and living-room play formats that gather households around co-present interaction [24]. Across these lines of work, wearables and related infrastructures function as expressive media that can be staged, shown, and circulated socially. However, less is known about how affective avatars on wearables specifically become shared objects of play through ambiguity.

2.3 Lived and Interpersonal Informatics

Personal informatics research has framed tracking as reflective practice rather than simple data accumulation [25]. Lived informatics highlights how tracking is embedded in routines, relationships, and creative repurposing [10, 26, 35]. Appropriation research similarly shows that people bend technologies toward unexpected ends [17, 42]. Importantly, tracking is rarely solitary: interpersonal informatics demonstrates how logs circulate within close ties and communities engage in collaborative interpretation [9, 30, 34]. Puusaar et al., for example, show that personal data is socially sensemade through collaborative interpretation and discussion [34]. However, that work tends to focus on making data understandable and actionable, whereas our findings show that, in this case, ambiguity itself becomes a resource for social interaction by opening a space for interpretation that people joke with, test, and negotiate together. These perspectives explain how data becomes conversational material, yet they rarely frame these practices explicitly as play forms organized through ambiguity. Building on this, a play lens explains how interpretation remains lightweight, relational, and repeatable rather than purely analytic. This leaves unclear how ambiguity becomes a resource that sustains playful, socially negotiated routines rather than accuracy-driven interpretation.

2.4 Stress Inference and Risks

Stress and emotion sensing stretch physiological proxies beyond what they can reliably infer, and in-the-wild use frequently involves mismatches between sensed signals and felt experience [8, 32]. Reviews of stress wearables highlight both potential benefits and persistent limitations [38]. Social designs that share biofeedback demonstrate how physiological signals can coordinate care and check-ins among close ties [21, 41]. Jiang et al.'s *IntimaSea*, for example, shows how shared stress displays can support awareness and relational care in close relationships [21]. However, such systems largely treat biosignals as a medium for communicating stress and coordinating support, whereas we show that ambiguity in the display can become a playable social mechanism rather than only a reflection or communication tool. At the same time, ambiguity does not automatically produce positive outcomes: it can become anxiety-inducing or socially risky when cues are treated as definitive verdicts. What remains missing is a clearer empirical account of when ambiguity supports playful coping versus when background uncertainty introduces harm, and what design safeguards help maintain supportive engagement.

Across these literatures, we know (1) biosignals can be used for communication or reflection, (2) personal data circulates socially, and (3) ambiguity can be treated as an interpretive or design property. What remains unclear is how an affective cue becomes a repeatable social play routine with recognizable loops (show → react → narrate → compare → care/test), and what safeguards keep such routines supportive rather than coercive. This paper addresses that gap by showing that ambiguous biosignal feedback is not only interpreted or shared, but in this case becomes a playable social mechanism through which users construct repeatable routines, and by surfacing the mechanics and boundary conditions that make ambiguity supportive versus harmful.

3 Method

3.1 Overview and Ethics

We conducted a multi-source qualitative study to examine how users interpret and socially appropriate an avatar-based stress indicator on a smartwatch (Figure 3). We triangulated three data sources—(1) App Store reviews, (2) public social media posts on RED (XiaoHongShu), and (3) semi-structured interviews—to capture both breadth (recurring sentiments and practices at scale) and depth (situated accounts of routines, meaning-making, and social dynamics). Institutional research ethics approval was obtained. All interview participants provided informed consent.

For public online content, we followed platform terms and established ethical guidance for internet research. We analyzed only publicly accessible reviews/posts, did not attempt to contact posters, removed usernames/handles and other identifying details, and quote sparingly. For non-English materials, we translated content into English for analysis and reporting and avoided including details that could enable re-identification.

3.2 Focal System: *StressWatch*

StressWatch is a third-party smartwatch application that estimates stress from heart-rate-derived signals (e.g., HRV and related measures available on the watch) and renders inferred states through a glanceable animated avatar on the watch face. Rather than presenting stress as a numeric score, the app conveys a small set of qualitative states via color and facial expression (e.g., green/calm, orange/elevated, red/high) and can issue brief prompts when the inferred state changes. The avatar is readily *showable* in co-present settings (via a wrist turn) and *shareable* at a distance (via screenshots).

We selected *StressWatch* as an instrumental case because it combines (1) an inference domain known to be noisy and context-sensitive (stress estimation from physiological proxies) with (2) a symbolic, socially legible representation (an expressive watch-face avatar) that facilitates rapid disclosure, circulation, and co-interpretation. We did *not* develop the app and do not assess clinical validity; our focus is on users’ interpretations, appropriation, and social practices around a contestable cue.

3.3 Data Sources and Collection

3.3.1 App Store Reviews. We collected user reviews from the Apple App Store, covering the period 2023–2025. Reviews provide concise, naturally occurring accounts of perceived value, breakdowns, and feature requests in everyday use. We exported review text and metadata (e.g., date, rating where available) and performed a two-stage screening.

Inclusion/exclusion. We included reviews that (i) referenced the stress avatar/watch-face display or stress states, and (ii) contained substantive experiential content (e.g., describing situations of use, perceived mismatch, sharing, routines, or impacts). We excluded (i) empty reviews and “rating-only” entries with no textual content, (ii) generic praise/complaints with no relation to the avatar or stress experience (e.g., “great app”), and (iii) clearly non-user content (e.g., spam or unrelated advertising). After screening, the final corpus included **238** reviews.

Translation for analysis. Reviews were written in multiple languages. To support consistent coding within a shared analytic frame, we translated non-English content into English prior to analysis. Of the 238 included reviews, 152 were originally in English, 71 were originally in Chinese, and 15 were in other languages (predominantly Japanese, Korean, Spanish, and German). Reviews originally in Chinese were translated by bilingual (English/Chinese) members of the research team. For reviews in the 15 other-language cases, we used machine translation (a current general-purpose neural MT system) to produce English versions for analysis, given the team’s language coverage. To reduce meaning drift, we (i) maintained a shared glossary for recurring terms (e.g., “stress,” “pressure,” “anxiety,” and HRV-related colloquialisms); (ii) spot-checked all 15 machine-translated reviews and a random 10% sample of the Chinese reviews—two researchers independently re-read each spot-checked item against the source, flagged passages where the English rendering felt unclear, idiomatic, or potentially distorting (e.g., culturally specific phrasing, sarcasm, or stress-related slang), and discussed flagged passages until reaching a shared interpretation; and (iii) retained the original-language text internally to support interpretation of excerpts when needed. Of the 15 machine-translated items, three were flagged and discussed; in two cases, the team adjusted

the working English gloss to better match the source, and in one case the item was excluded from quoted excerpts (though retained for thematic coding) because re-identification risk and translation uncertainty together made quotation unsafe. In reporting, we present English translations and omit details that could enable re-identification.

3.3.2 Public Social Media Posts (RED/XiaoHongShu). To examine social circulation and collective interpretation, we collected publicly accessible posts from RED (XiaoHongShu) in 2024–2025 using Chinese keywords for “stress monitoring” and the app name. We focused on RED because it supported a dense ecology of public screenshot sharing and comment-thread discussion around interpretation and sharing practices.

Inclusion/exclusion. We retained posts that met at least one of the following criteria: (i) included a screenshot/photo of the watch-face avatar, (ii) described first-person use (routines, incidents, interpretations, or effects), and/or (iii) contained comment-thread discussion about how to read, share, or respond to the avatar. We excluded (i) posts that mentioned stress tracking without referencing the avatar/watch-face display, (ii) repost-only content with no added interpretation, (iii) apparent promotions/advertisements or influencer marketing without first-person use detail, and (iv) posts with insufficient context to interpret (e.g., image-only with no caption and no discussion). We retained associated comment threads when present because they often contained peer calibration and “rules of thumb.” The final corpus comprised **N=174** posts (including associated comment threads when present). All RED posts and comments were originally in Chinese and were translated by bilingual members of the research team using the same glossary and spot-checking protocol described above; no machine translation was used for this corpus. Handles and identifying details were removed.

3.3.3 Interviews. We conducted semi-structured interviews with **18** *StressWatch* users recruited via social media and snowball sampling. Inclusion required at least one month of use to ensure participants could speak to routine interaction and appropriation over time. Participants (ages **22–41**, **9 women/9 men**) were located primarily in **North America (n=10)** and **China (n=6)**, with **two supplementary interviews in Europe (n=2)**. Interviews lasted **45–60 minutes**, were conducted via video or phone, audio-recorded with consent, and transcribed. Table 1 summarizes participant demographics, location, duration of *StressWatch* use, recruitment source, and interview mode.

The interview protocol covered: (1) everyday routines (when/why participants checked the avatar), (2) memorable incidents (humorous, tense, or socially consequential moments), (3) perceived mismatch and trust calibration, (4) responses to prompts/alerts, (5) sharing practices (co-present showing, screenshots, group rituals), and (6) perceived impacts on wellbeing, relationships, and work/study contexts. Representative questions for each topic are reported in Appendix A.

Note on analytic emphasis. While all interviews informed our thematic analysis, we treat North America and China as the primary analytic contexts because they reflect distinct sharing ecologies (public posting/commenting on RED versus more private, messaging-based circulation). The two supplementary interviews from Europe are included to broaden the range of experiences but are not used to substantiate cross-context comparisons.

3.4 Analysis

Analytic process. We conducted Reflexive Thematic Analysis (RTA) across all three sources following Braun and Clarke’s phases [4]. Analysis was iterative and interpretive rather than aimed at producing a fixed, exhaustive codebook. While play informed our analytic lens, the five play forms were iteratively identified through the data and refined through comparison across sources

Table 1. Interview participants (N=18). Duration of use is at the time of interview. Recruitment source: SM = social media (X/Twitter, Reddit, RED); SB = snowball referral from a prior participant. Mode: V = video call; P = phone.

ID	Age	Gender	Location	Occupation	Duration of use	Recruitment	Mode
P1	25	M	North America	Graduate student	~3 months	SM	V
P2	22	M	China	Undergraduate student	~2 months	SM	V
P3	26	F	North America	Designer	~6 months	SM	V
P4	22	M	China	Undergraduate student	~4 months	SB	V
P5	24	F	North America	Software engineer	~3 months	SM	V
P6	30	F	North America	Healthcare worker	~8 months	SM	V
P7	29	M	China	Product manager	~5 months	SB	V
P8	28	F	North America	Consultant	~7 months	SM	V
P9	27	F	North America	Marketing	~4 months	SB	V
P10	32	F	China	Researcher	~6 months	SM	V
P11	31	M	North America	Engineer	~5 months	SM	V
P12	34	F	China	Teacher	~3 months	SB	P
P13	22	F	China	Graduate student	~2 months	SM	V
P14	40	M	North America	Manager	~10 months	SM	V
P15	30	M	North America	Writer	~4 months	SB	V
P16	41	M	Europe	Analyst	~6 months	SM	V
P17	27	F	North America	UX researcher	~5 months	SM	V
P18	28	M	Europe	Engineer	~3 months	SB	V

rather than imposed in advance. First, three researchers familiarized themselves with each corpus and generated initial open codes on an initial subset of interview transcripts and a stratified sample of reviews/posts (e.g., *Perceived mismatch*, *Showability*, *Screenshot sharing*, *Humor/memes*, *Care/companionship*, *Competitive/cooperative rituals*, *Worry spirals*, *Audience risk*). For initial coding, we selected stratified samples to capture variation in tone and use contexts (e.g., high/low star ratings and early/late periods for reviews; posts with/without screenshots and with/without substantial comment threads on RED). We then held regular analytic conversations to compare interpretations, surface assumptions, and refine the analytic focus, documenting decisions through reflexive memos. Rather than seeking coder “agreement,” we used difference as an analytic resource: points of tension (e.g., whether a practice was primarily “play” versus “coping,” or “social signaling” versus “pressure”) were traced back to data excerpts and used to sharpen theme boundaries and articulate boundary conditions. Coding continued alongside ongoing theme development through constant comparison within and across sources, culminating in a thematic schema that structures our Findings.

Source-specific contributions. We approached the three corpora as complementary lenses on the same phenomenon and treated their distinct affordances explicitly. App Store reviews contributed concise, naturally occurring accounts of perceived value, breakdowns, and recurring complaints; they were especially useful for surfacing the prevalence of certain framings (e.g., “mood badge,” “my watch is dramatic”) and for capturing dissatisfaction that interview participants may have been less likely to volunteer. RED posts and their comment threads contributed visible, in-progress social interpretation: screenshot sharing, peer calibration (“is it just me?”), folk theories accreted across replies, and meme-able framings that travelled beyond individual users. Interviews contributed rich situated accounts—routines, incidents, contextual reasoning, and trust calibration over time—that helped explain why the short fragments seen in reviews and posts arose in the first place. We first generated initial codes within each corpus to preserve source-specific meanings, then developed candidate themes through cross-corpus comparison, retaining a theme only when it was supported by multiple sources or when we could clearly justify it as source-specific (and label it as such). We used a shared theme workspace to map how evidence accumulated across corpora and to avoid interviews being treated as the default “ground truth.”

Negative cases and boundary conditions. Consistent with an RTA commitment to nuance and complexity, we actively attended to accounts that complicated an emerging “play” interpretation, including: users who treated the avatar as diagnostic/authoritative, users who avoided sharing or hid the watch face in public/work contexts, users who disengaged after mismatch, and users who experienced monitoring as burdensome or anxiety-inducing. We used these cases to articulate boundary conditions for each play form and to ground the failure modes reported in our Discussion (e.g., context collapse, worry spirals under persistent “red,” and moralization through comparison).

Quality and transparency. We attended to quality through (i) coherence between themes and supporting extracts, (ii) depth by examining variation and boundary conditions, and (iii) transparency via a reflexive audit trail (memos and meeting notes) capturing analytic decisions and shifts in interpretation. Triangulation across corpora was used to enrich interpretation and test whether themes traveled across contexts, not to “validate” a single ground truth account.

Researcher reflexivity. Our analysis was shaped by our backgrounds in HCI and play-oriented interaction research, which sensitized us to ludic framings of everyday practices. We used reflexive memos to surface how this lens influenced coding decisions (e.g., when labeling an interaction as “play” versus “coping”), and we revisited excerpts that initially appeared playful to examine whether participants framed them as voluntary, low-stakes, and socially negotiated. Specifically, when we initially coded an excerpt as play, we revisited it against three criteria: whether participants framed the practice as voluntary rather than required, low-stakes and reversible, and socially negotiated rather than individually imposed. Excerpts that did not meet at least two of these criteria were recoded under alternative themes such as coping, social signaling, compliance, or burden. This check helped us distinguish playful appropriation from practices that were merely stressful, obligatory, or instrumental.

4 Findings

Across interviews, App Store reviews, and posts on RED, the smartwatch stress avatar was frequently treated as more than an individual, clinical indicator (Figure 4). Rather than interpreting it only as self-tracking feedback, people repeatedly played with it: performing states, inventing rules, telling stories, and caring for the character. Across the five forms below, we see participants acting on the playable gap between system output and lived experience: rather than treating mismatch as a reason to disengage, they used it as a prompt to interpret, joke, compare, and probe. In our materials, perceived mismatch was a recurring theme yet rarely the end of engagement; participants who described persistent mismatch frequently described continued use through the play forms below, often citing the avatar’s social legibility and the small rituals it had come to anchor as reasons to keep it on. We therefore organize our findings as five forms of everyday play around an affective wearable companion. We attribute interview excerpts to participants (P#) and present other excerpts as *App Store review* or *RED post*.

RQ-to-Findings map. To make our answers to the research questions traceable, Table 2 maps each play form to (RQ1) how participants *interpret, share, and use* the avatar-based cue in everyday life when it remains open to interpretation, and (RQ2) which mechanics (visibility, contestability, circulation, persistence) enable the routine, along with boundary conditions that keep it supportive and failure modes that make it harmful.

4.1 Themes

The final schema structures our Findings into five forms of play:

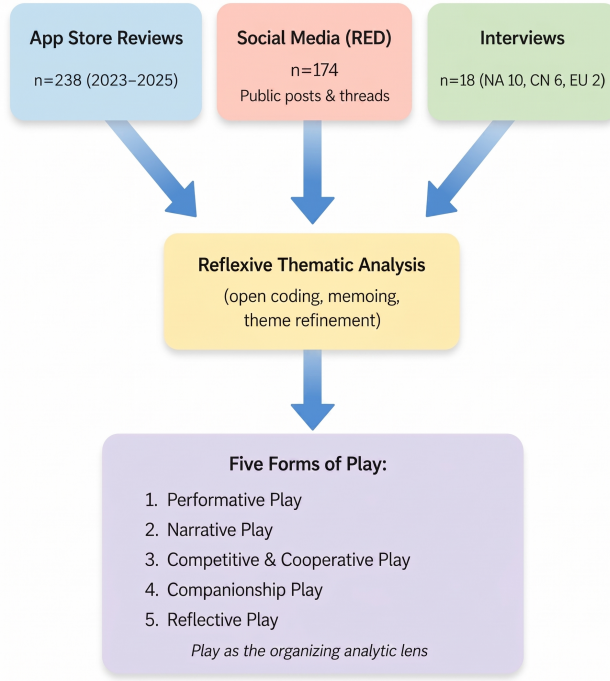


Fig. 3. Study design schematic illustrating our three data sources (238 App Store reviews collected 2023–2025; 174 RED/XiaoHongShu posts collected in 2024–2025; and 18 semi-structured interviews with participants in North America (n=10), China (n=6), and Europe (n=2)), the reflexive thematic analysis process, and the five forms of play that structure our findings.

- **Performative Play:** Using the avatar as a social signal—showing, sharing, and staging states through wrist gestures and screenshots.
- **Narrative Play:** Making sense through stories—jokes, folk theories, and collective world-building around mismatches.
- **Competitive & Cooperative Play:** Games with stakes—low-stakes competitions, solidarity practices, and coordinated rituals.
- **Companionship Play:** Treating the avatar as a pet—personification, care behaviors, and soothing rituals.
- **Reflective Play:** Puzzles and experiments—pattern-spotting, micro-experiments, and playful inquiry into “why it went red.”

Together, this multi-source design captures both large-scale discourse (reviews and public posts) and rich, situated accounts (interviews), enabling a comprehensive view of how an “imperfect” stress avatar becomes socially usable through playful appropriation.

4.2 Performative Play: The Avatar as Social Signal

Across sources, interviewees and reviewers/posters described routine moments where the avatar was made visible to others (e.g., by raising a wrist during co-present interaction) and later shared at a distance via screenshots. We frame these practices as performative play: participants used the companion as a stage prop to control timing, tone, and audience, experimenting with how much to reveal while keeping the interaction lightweight. In these accounts, the avatar functioned as

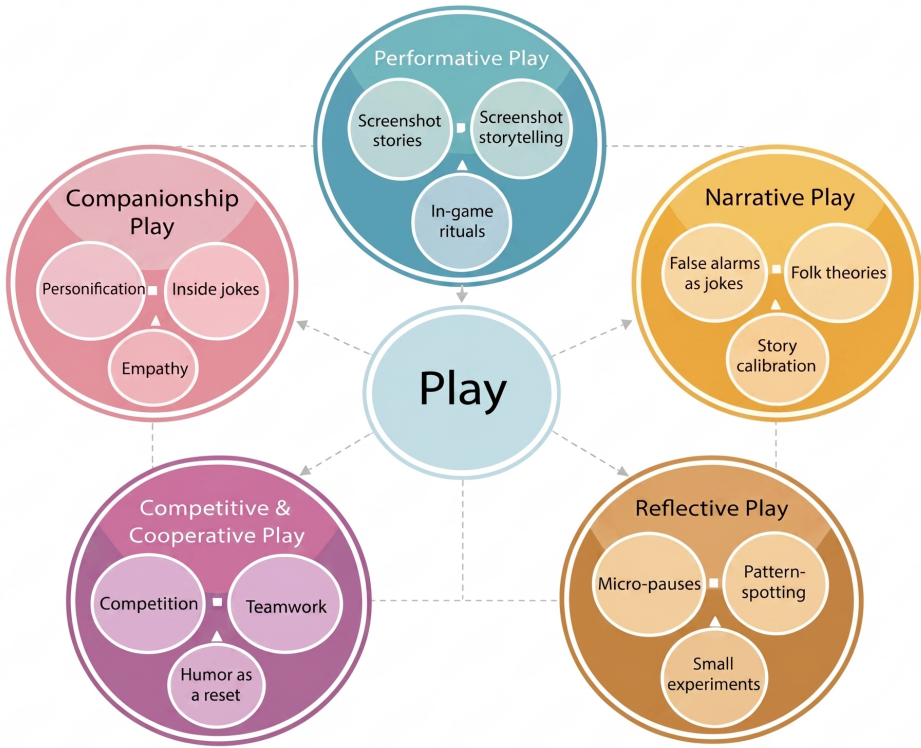


Fig. 4. Thematic map showing five forms of play around an affective wearable companion. Play serves as the organizing framework, with each form representing distinct but interconnected playful practices. Dashed lines indicate how play forms support and enable each other (e.g., performative play provides material for narrative play; competitive and cooperative play creates occasions for companionship).

a playful, non-verbal cue that could prompt practical responses (e.g., offers of help, changes in conversational tone, or coordination decisions about chores and plans).

4.2.1 Wrist-as-status as performance. Participants routinely showed the avatar instead of explaining themselves. P3 (26F, North America) put it plainly:

“When I’m too drained to talk, I just raise my wrist and show the grumpy face. It says, ‘rough day,’ without me saying anything.”

Table 2. Mapping research questions to findings. For RQ1, we show how participants *interpret, share, and use* the avatar-based stress cue. For RQ2, we identify the enabling mechanics and the boundary conditions/failure modes that make routines supportive versus harmful.

Play form (Findings)	RQ1: Interpret / Share / Use (what people do)	RQ2: Mechanics & boundary conditions (supportive vs harmful)		
		Primary mechanics	Supportive when... (boundary conditions)	Harmful when... (failure modes)
Performative play (avatar as social signal)	Share: wrist-turn showing; screenshot check-ins. Use: lightweight disclosure and coordination (tone-shifting, help, chore/plan renegotiation). Interpret: treated as a cue, not an explanation, enabling “say it without saying it.”	Visibility (glanceable on watch face) Circulation (screenshots)	Audience is trusted and norms frame the avatar as <i>non-diagnostic</i> and <i>optional</i> ; users can choose <i>timing</i> and <i>audience</i> for showing; co-present reading is playful/empathetic rather than evaluative.	Context collapse (work/authority audiences) creates compelled disclosure; observers treat the cue as a verdict; visibility becomes mis-timed or uncontrollable, producing embarrassment, pressure to explain, or misinterpretation.
Narrative play (stories, jokes, folk theories)	Interpret: mismatch becomes a prompt for humor, memes, nicknames, and folk theories (e.g., “it only sees stillness”). Share: stories circulate through posts/comments and group chats; peer calibration (“is it just me?”). Use: stabilizes group “rules of thumb,” reducing stakes and normalizing uncertainty.	Contestability (qualitative, mismatch-prone cue) Circulation (threads/memes)	Communities provide interpretive resources (peer reassurance, shared heuristics); ambiguity is framed as <i>playable</i> and <i>non-alarming</i> ; users maintain emotional distance from the reading and can laugh off “glitches.”	Uncertainty burden turns into worry spirals when users lack scaffolding or when “red” persists without explanation; ambiguity becomes anxiety-inducing rather than playful; misinformation/over-interpretation spreads through folk theories.
Competitive & cooperative play (games with stakes)	Use: negotiated rules and low-stakes consequences (coffee/boba/chores; time-boxed challenges); solidarity rituals (“we’re all red today”). Share: comparisons co-present or via screenshots. Interpret: the avatar becomes a “score” that is socially negotiated (teasing vs care).	Visibility (easy comparison) Persistence (states endure long enough to “count”) Circulation (sharing)	Stakes stay small and reversible; participation is opt-in; group norms emphasize support over performance; competition flips to care when someone stays “red,” enabling gentle check-ins and coordinated breaks.	Moralization/pressure emerges when comparisons become evaluative (stress as personal failure), when workplace productivity norms leak in, or when users feel obligated to “perform calm”; persistent “losing” fuels shame or withdrawal.
Companionship play (avatar as pet/character)	Interpret: personification reframes the reading as the character’s mood (emotional distance from “diagnosis”). Use: soothing rituals (“calm it down, which calms me down”); softer self-talk. Share: nicknames and character talk make disclosure indirect and socially legible.	Persistence (ongoing companion) Visibility (always-present character)	Users can downshift intensity and treat the avatar as <i>exaggerated</i> rather than authoritative; interaction stays lightweight (brief soothing actions, not constant checking); social references are playful rather than diagnostic.	A “needy/accusatory companion” emerges when negative states persist, prompting repeated checking and irritation; the character becomes a source of obligation or blame; users feel watched by others or by the device.
Reflective play (puzzles, experiments, patterns)	Interpret: “why did it go red?” as casual inquiry; triangulation with routines (meetings, caffeine, commuting). Use: micro-pauses, habit tweaks, personal “rules of thumb,” low-pressure self-experimentation. Share: comparing notes with peers; exchanging patterns and explanations.	Contestability (invites hypothesis) Persistence (supports pattern recognition)	Users treat the cue as a <i>prompt</i> rather than truth; inquiry is time-boxed and low-stakes; there are off-ramps (pause, ignore, reframe) that prevent rumination; social support normalizes uncertainty.	Over-monitoring/rumination when repeated checking replaces coping; spurious causal beliefs harden; persistent “red” plus contestability creates self-surveillance spirals, especially without explain-on-demand context.

The effect of this compact display was immediate and practical: a partner took over chores; a roommate softened their ask; a colleague stepped in. P15 (30M, North America) described the ritual at home:

“My roommate and I show each other our bubbles at night. It’s like saying ‘how are you’ without words.”

Reviews converged on the same function—*“It’s like wearing a mood badge—I don’t have to explain myself at the end of a long day.”*—while posts on the Chinese platform staged the moment more theatrically (two wrists, same red face; captions of solidarity). In both registers, a private measure was converted into a legible cue that others could act on, compressing explanation into a glance while preserving dignity and pace.

4.2.2 Screenshot storytelling as playful narrative. When people were apart, the avatar traveled as a picture. A single screenshot after a difficult meeting replaced a paragraph of venting. P17 (27F, North America) explained the efficiency:

“I send screenshots to my partner after bad meetings. One picture says it all.”

Couples used nightly comparisons as a shorthand debrief; students posted late-night reds framed as “deadline energy.” P7 (29M, China) described the cadence:

“Every night, we compare our stress bubbles—like sharing how our day went, through the little faces on our watches.”

Reviews named the communicative texture—*“I send the faces to my group chat. It’s funnier than typing stressed.”*—and replies often layered humor or encouragement. The screenshot thus functioned as a story fragment that invited response and made stress collectively discussable without demanding formal disclosure.

4.2.3 Ritualized play for group coordination. Beyond single moments, small, repeatable practices formed around the avatar. In our interviews, some participants described workplace or friend-group rituals: checking colors before an afternoon break; whoever was deepest red “owed” coffee; and if someone stayed red all week colleagues opened a check-in. P11 (31M, North America) summarized one such office game:

“Afternoon break, we tap watches. Whoever’s in the red buys coffee. If someone’s red all week, we ask what’s up.”

At home, families compared bubbles before dinner to decide who cooked; among students, pre-exam comparisons calibrated pep talks and plans (“We all showed red today. At least we suffer together.”). Light stakes—snacks, turns, chores—kept attention gentle while encoding a norm: when the face stays dark, the group adjusts.

4.2.4 Visibility as a contested play mechanic. The same visibility sometimes misfired. P5 (24F, North America) recalled a meeting where the avatar surfaced at the wrong time:

“In a meeting, the orange face lit up. My manager asked, ‘Everything okay?’ I wasn’t—it was just too much coffee.”

Some dimmed screens or hid complications on demanding days; reviewers complained that *“The red guy shows up at the worst times—wish I could hide it.”* Across reviews and posts, friends occasionally used visible timestamps to “call out” all-nighters. The avatar’s ambient presence thus created a double edge: it invited empathy and accountability, but it could also expose private states or mislead observers when context shifted.

Boundary conditions and risks. Performative play depended on a shared understanding that the avatar was a lightweight cue rather than a definitive account of someone’s internal state. When ambiguity was treated as a verdict (or read as a performance obligation), participants described

feeling put on the spot, pressured to explain, or worried about being misread. Unwanted visibility—especially in workplaces or around acquaintances—could turn the same mechanic that enabled humor and care into a source of social risk.

Theme synthesis. Across sources, the avatar was routinely used beyond individual self-monitoring: it was made visible in co-present settings, shared via screenshots, and incorporated into small group practices (e.g., check-ins, coordination, and low-stakes games). In these cases, participants emphasized the avatar’s communicative and coordination value, even when they questioned the accuracy of the underlying stress inference.

4.3 Narrative Play: Making Sense Through Stories

Interviewees and users frequently reported mismatches between the avatar state and their felt experience (e.g., “high stress” during leisure, or “calm” during anxiety). Rather than ending use, participants treated this uncertainty as a prompt for narrative play: telling jokes, inventing folk theories, and recruiting peers as co-authors of what the companion “really means” in context.

4.3.1 Jokes, memes, and “glitches” as prompts. Many households had names for the behavior. P9 (27F, North America) described how a misfire became a character in the living room:

“I was on the couch watching a comedy. Ding—‘High Stress!’ We now call it my dramatic Tamagotchi. My partner says, ‘Your pet is having a meltdown again.’”

Friend groups anticipated specific contexts. P2 (22M, China) laughed about gaming:

“When I played games with friends, the bubble always turned red. They would laugh and say my watch is the real loser. It was never serious—it made our group chat funnier.”

Reviews echoed the pattern—“*It freaks out during yoga, which is hilarious—I send the screenshot to my family.*”—and social posts remixed karaoke reds into lighthearted memes (“Apparently singing stresses me out more than exams”). Through repetition, error became predictable enough to be domesticated into humor, keeping the artifact in circulation.

4.3.2 Folk theories as collective worldbuilding. The inverse mismatch—calm face, stressed person—prompted explanation rather than dismissal. P1 (25M, North America) reasoned through public speaking:

“Green smile before a big talk—palms sweaty! I figured it only ‘sees’ stillness, not my head.”

P13 (22F, China) arrived at a similar account before an exam:

“I was so nervous, but my watch was green. I told my friends: maybe it can’t feel nerves, only heartbeat.”

Reviews wrestled with analogous dissonance—“*It told me I was calm while I had a panic attack. Guess it doesn’t know me.*”—while comment threads accumulated hypotheses about breathing, posture, caffeine, movement, and algorithmic thresholds. Individually tentative ideas stabilized into shared “rules of thumb” that users applied in future interpretations. Here, ambiguity worked as a resource: it opened space for collective interpretation rather than closing interaction down.

4.3.3 When ludic ambiguity stops feeling safe. For some, mismatches provoked worry. P12 (34F, China) recounted a spiral:

“It showed ‘high stress’ for hours while I felt fine. I spiraled—what if something’s wrong?”

She later found a benign cause—“*T’d had wine. That explained it.*”—but the intervening uncertainty carried a cost. P16 (41M, Europe) described a similar loop:

“I Googled high HRV stress for two hours after my watch went red. It scared me more than my actual stress.”

Reviews captured the same affect—*“It said I was stressed all night, now I wonder if my heart is okay.”*—pointing to an uneven distribution of ambiguity’s burden: without interpretive resources, uncertainty can amplify rather than soften concern.

4.3.4 Peer calibration as collaborative sensemaking. Across data sources, users turned to others to normalize oddities. Threads that began with “why red during gaming?” drew dozens of replies; review Q&A exchanges accumulated reassurance; interviewees described searching first before changing habits. P7 (29M, China) summarized the practice:

“I always look on RED to see if it’s just me. If many people post the same, I feel okay.”

Calibration redistributed expertise across communities and stabilized expectations: if others’ watches also “freaked out” in karaoke, then the device—and by extension, the user—was not broken.

Theme synthesis. Across sources, perceived mismatch did not simply end use. Instead, ambiguity shifted effort to interpretation: users relied on humor, informal “rules of thumb,” and peer discussion (e.g., review Q&A and RED threads) to contextualize signals and reduce uncertainty. This shows how ambiguity, rather than accuracy alone, organized social play around the cue.

4.4 Competitive & Cooperative Play: Games with Stakes

Here, the avatar becomes a game piece. Participants did not merely share readings; they negotiated rules (who “wins”), invented light consequences (coffee, boba, chores), and used comparison as a cooperative practice for solidarity and care. This is not points-based gamification imposed by a system, but everyday, socially negotiated play where the “score” is an expressive character state that can be teased, protected, or collectively managed.

4.4.1 Low-stakes scorekeeping and friendly rivalry. Participants layered simple games onto ordinary days. P4 (22M, China) described a weekly rule:

“Weekly challenge—lowest stress wins. Highest stress buys boba.”

Couples ran time-boxed challenges (“stay green till lunch”); P18 (28M, Europe) compared commutes:

“We make it a game—who can stay green longest during the workday.”

Parents gamified bedtime calmness with children; reviews celebrated the format—*“My kids compete with me for who’s calmest at bedtime.”* The shared scoreboard made micro-regulation visible enough to act on without turning it into moral judgment. People checked more often, tried breaths together, or adjusted task order to protect a streak.

4.4.2 Solidarity and “suffer together” play. Beyond rivalry, cooperative play took the form of shared acknowledgment: noticing collective “red moments,” validating bad days, and treating stress as something the group could face together rather than individual failure. P2 (22M, China) framed it as camaraderie:

“When everyone’s bubble is red, we just laugh—like okay, we’re all suffering together.”

In these moments, the game shifted from competition to mutual care. People traded screenshots as small check-ins, offered quick comfort (“take a break”), or synchronized micro-actions (breathing, water, a short walk) to help the group “cool down.” Competitive framing could flip into care when someone’s avatar stayed red for too long.

Boundary conditions and risks. Although competitions and comparisons were often framed as playful, they sometimes introduced subtle pressure. Participants described moments where repeated “red” states felt like losing, or where comparisons risked moralizing stress as an individual responsibility rather than a shared condition. In workplace contexts especially, playful challenges could blur into expectations about productivity or emotional control. Some participants responded by opting out of group comparisons or reframing challenges as cooperative rituals rather than competitions.

4.5 Companionship Play: The Avatar as Pet

Separate from games with stakes, many participants treated the watch-face character as a pet-like presence: something to name, talk about, and gently manage. This companionship framing recast stress readings as a character’s moods—reducing verdict-like pressure while still creating a prompt for attention and small soothing actions.

4.5.1 Personification and nicknames. Nicknames such as “drama queen,” “worrier,” “pet bubble,” and “Tamagotchi” were not simply decorative. They reframed the reading as a socially interpretable character trait (“the watch is being dramatic”) rather than a clinical verdict about the user. Personification also made the avatar easier to reference in conversation (“it’s mad again”) and easier to share without over-disclosing (“my pet is acting up”).

4.5.2 Care behaviors and soothing rituals. The avatar was sometimes treated as an entity to respond to or “take care of.” P6 (30F, North America) put the reflex simply:

“If it’s red, I ‘calm it down’—which calms me down.”

This framing made it easier to respond with small, low-cost coping actions (e.g., taking a walk, drinking water, or stretching) without treating the reading as definitive. Care took the form of tiny, repeatable rituals: checking in, doing a brief reset, and watching for the character to “settle.”

4.5.3 When the companion “misbehaves.” Companionship was not always comforting. When the avatar stayed red during calm moments, or flipped unpredictably, users described irritation and worry: the “pet” felt needy, dramatic, or accusatory. Some responded by hiding the complication, disabling the watch face, or reinterpreting the character as comic exaggeration rather than truth. These moments clarify a boundary condition for companionship play: the same personification that softens stakes can also amplify attention when the companion is persistently “upset.”

Boundary conditions and risks. Companionship play relied on users maintaining emotional distance: treating the avatar as a character rather than a diagnosis. When the avatar stayed “upset” for long periods, some participants felt blamed or pulled into repeated checking, and a few described irritation or worry that made the companion feel needy rather than supportive. These cases suggest companionship works best when participants can downshift visibility, reinterpret the character as exaggerated, or temporarily disengage without losing face in social settings.

Theme synthesis. Play did not trivialize stress; it made regulation livable. By recasting signals as scores, mascots, and jokes, people created small pressures and small permissions that sustained attention without heaviness. The same features that fed games also fed care.

4.6 Reflective Play: Puzzles, Experiments, and Patterns

Even when participants doubted readings, many treated the avatar as a puzzle to solve or a game to understand. “Why did it go red?” became a playful investigation—testing theories (coffee? stress? movement?), running micro-experiments, and comparing notes with others. Over time, these small

inquiries accumulated into pattern-spotting that felt less like clinical self-tracking and more like detective work or personal science play [23].

Boundary conditions and risks. Reflective inquiry was not always beneficial. For some participants, repeated checking and pattern-spotting risked turning into rumination or over-monitoring, particularly when signals felt inconsistent. Rather than supporting insight, ambiguity could encourage excessive self-surveillance or misplaced causal assumptions. Participants mitigated this by treating the avatar as a loose prompt rather than a diagnostic tool, or by temporarily disengaging when monitoring felt burdensome.

4.6.1 Micro-pauses as playful resets. Participants valued the nudge precisely because it arrived at unplanned moments. P10 (32F, China) described a pragmatic routine:

“Even if it’s off, I use it to breathe for a minute.”

Reviews echoed the tone—“*I don’t know if it’s right, but it reminds me to slow down.*”—and people reported being grateful for the excuse to momentarily step out of pace without justification.

4.6.2 Pattern-spotting as puzzle-solving. With repeated exposure, spikes linked to contexts: recurring meetings, commutes, caffeine, late-night screens, exam weeks. P8 (28F, North America) translated noticing into preparation:

“Every Tuesday afternoon goes red—my team meeting. Now I prep better.”

Posts shared week-long “streak” collages to reflect on what changed when sleep or steps improved; reviewers mentioned predictable spikes after scrolling in bed. Diffuse stress was reframed as time- and event-linked, which made it manageable.

4.6.3 Experiments as games with yourself. Participants tried adjustments whose effects could be watched: cutting afternoon coffee, short walks, moving a difficult task, limiting doomscrolling. P14 (40M, North America) noticed a simple lever:

“Cutting afternoon coffee stopped the post-3pm red streaks.”

A post celebrated a similar tweak: “After daily walks, my bubble finally stayed green.” Micro-games and group rituals doubled as accountability, making experiments social without being punitive.

4.6.4 Critical engagement and triangulation. Skepticism and value coexisted. P1 (25M, North America) captured the stance:

“I don’t believe the number, but I believe paying attention helps—and the app keeps me doing that.”

Some kept short notes; others cross-checked sleep or steps; many searched HRV explainers. The point was not to ratify the device but to calibrate expectations and keep what worked.

Theme synthesis. Reflection emerged as an outcome of play rather than a prerequisite. Alerts created occasions; social circulation created momentum; patterns and experiments followed. The practice thrived not because the avatar was precise but because it was glanceable, shareable, and repeatable.

4.7 Inter-theme Dynamics: How Signals Become Practices

Taken together, the five play forms show how a glanceable avatar becomes embedded in everyday practice. Participants performed states for others (Performative Play). When signals were mismatched, they built stories, folk theories, and reassurance through shared interpretation (Narrative Play). Social circulation then enabled socially negotiated games with small stakes and cooperative rituals (Competitive & Cooperative Play), including solidarity-oriented “suffer together” practices.

Alongside (and sometimes in tension with) these games, participants also engaged in pet-like companionship with the character (Companionship Play). Over time, repeated engagement supported pattern-spotting and micro-experiments framed as low-pressure inquiry (Reflective Play). Overall, the avatar’s value in our data depended less on measurement precision than on how it functioned as a playable social artifact that supported interpretation and action in context.

5 Discussion

In this discussion, we develop four arguments. First, we situate ambiguity-sustained play against ordinary gamification, showing that interpretive flexibility—rather than score optimization—explains the social practices we observed (§5.1). Second, we describe four design mechanics that organize how ambiguity becomes playable in this case (§5.2). Third, we identify when these mechanics break down and produce harm (§5.3). Finally, we delimit the contribution to StressWatch’s specific representational form rather than to all biosignal wearables (§5.4).

This framing clarifies what the playable gap contributes theoretically. Prior HCI and design work shows that ambiguity can be a productive interpretive resource rather than only a usability problem. StressWatch extends that view by showing how, in this instrumental case, ambiguity can organize repeatable social play loops around biosignal feedback when cues are visible, circulatable, persistent, and contestable. It also complicates accuracy-centered accounts of affective wearables: perceived mismatch did not always end engagement, but sometimes became material for joking, comparing, caring, and reflecting. This does not make inaccuracy desirable; rather, contestability may support socially negotiated play when the cue is not treated as diagnostic truth.

5.1 What Ambiguity Adds Beyond Ordinary Gamification

A common explanation for play around health technologies is gamification: scores, ranks, streaks, and rewards layered on top of measurement to drive engagement [5, 22]. Precise-metric systems—step counts, calorie totals, fitness ranks like WeRun—typically work by stabilizing what counts as a “score,” so that comparison and competition are organized around fixed values. Our data show a different logic at work. *StressWatch* does not present a ranked score; it presents a qualitative, contestable cue whose meaning is not fixed by the system. Comparison still happens, but it is comparison *across interpretations*: groups negotiate whether a red face means “stressed,” “too much coffee,” or “the watch is being dramatic.”

This matters because it changes what comparison *does*. In score-based gamification, ambiguity is friction: it makes leaderboards unreliable and reduces the legibility of competition. Here, ambiguity is the resource that makes the cue socially playable: it lets users joke, recalibrate, and flip competition into solidarity (“we’re all red today”) without committing to a strict ranking. Where step-count rivalries depend on precision, the rivalries we observed depended on *interpretive flexibility*—the freedom to read the same red face as funny in one moment and worth a check-in in another. This is what ambiguity adds beyond ordinary gamification: not engagement-by-measurement, but engagement-by-interpretation, where social meaning, not the score, is the primary product. It also helps explain why perceived mismatch did not straightforwardly predict abandonment in our materials: contestability kept the cue conversational rather than authoritative.

5.2 Four Mechanics of Ambiguity-Sustained Play

Rather than attributing outcomes to abstract “playfulness,” our data point to concrete mechanics that enabled and constrained appropriation. These mechanics name the design surface through which ambiguity becomes socially actionable.

The playable gap appears most strongly when three conditions overlap: the system output is visible and emotionally legible, the user can compare the output with their own felt experience,

and the output can be shared or interpreted socially. In StressWatch, the avatar's simple emotional expression, delayed or imperfect biosignal inference, and screenshot-friendly watch-face design together create opportunities for users to joke, test, compare, and narrate the gap between measurement and experience. Ambiguity becomes playable under social conditions where others treat the cue as low-stakes and contestable, especially among trusted ties; when it is read as a verdict or exposed in power-asymmetric settings, the same gap can become burdensome rather than playful.

Visibility. Because the avatar sits on the watch face, it is legible at a glance and easy to surface in co-present interaction via a wrist turn. This helps explain why Performative Play appears so frequently: the cue is already “on stage.” Visibility can coordinate care (partners shifting chores, friends checking in) but also create risk (a manager noticing an orange face at the wrong moment). It is therefore not a neutral affordance; it sets the social conditions under which the signal is interpreted.

Contestability. Users treated the avatar as contestable because it was qualitative, symbolic, and often mismatched with felt experience. Contestability fuels Narrative and Reflective Play: when the cue can be challenged, users ask “why” and generate hypotheses, creating space for interpretation and humor [2, 14]. As discussed below, this includes both temporal latency and algorithmic error. Contestability also has a cost: without interpretive resources, uncertainty can tip into worry, as when prolonged red is read as hidden health trouble.

Circulation. Screenshots are a low-friction way to move the cue across distance. Circulation supports multiple play forms at once: performative check-ins (“one picture says it all”), narrative remixing (memes), and peer calibration (asking others if a mismatch is normal). These dynamics resemble social sensemaking in interpersonal informatics [34], but the avatar's character-like form makes circulation feel closer to sending an emoji or sticker than sending data, lowering the barrier to sharing and the perceived stakes of disclosure. This is part of why StressWatch differed from precise-metric systems in our data: a fixed score rewards ranking, while an expressive character rewards remixing.

Persistence. Because the avatar updates over time and can remain in a state, it supports repeated engagement (daily comparisons, routines, streak-like narratives). Persistence enables companionship (a “pet” that can be soothed) but also drives failure modes (a “needy” companion, repeated checking, spirals when red persists). The same persistence that builds attachment can also amplify burden, particularly when the system offers little explanation for why a state persists.

Taken together, these four mechanics explain why this expressive avatar on the wrist could function as an ambiguity-sustained ludic system in everyday life. Each mechanic is also a lever for mitigation: visibility can be controlled, contestability scaffolded, circulation made consensual, and persistence downshifted.

5.2.1 When Mismatch Is Delay, Not Error. Not all perceived mismatch has the same status. Algorithmic error occurs when a reading feels ungrounded or simply wrong: the avatar reports high stress in a context users cannot connect to bodily state, activity, or recent experience. Temporal latency is different: the reading may correspond to an earlier event, but arrives late enough that users interpret it against the wrong moment. Interpretive openness is different again: the reading is plausible but underdetermined, leaving room for several explanations. This distinction matters for play because delayed cues can still be folded into stories, jokes, or retrospective interpretation: “maybe it is reacting to the meeting I just left” keeps the signal usable as conversational material and sustains playful negotiation. Clearly wrong algorithmic readings more often disrupt trust and collapse play into dismissal or worry. A design implication is to surface timing without eliminating

ambiguity. For example, a coarse “inferred in the last 10–20 minutes” or “when this was inferred” cue could help users reason about whether the avatar is responding to a current state or a recent event, while preserving the interpretive flexibility that makes the cue socially negotiable.

5.3 When Play Breaks Down

Playful appropriation is conditional. Across sources, play stayed lightweight when the avatar was treated as a cue among trusted ties and interpreted as “not a verdict.” When those conditions broke, the same mechanics produced predictable harms. Naming these failure modes connects design safeguards to concrete risks.

Context collapse. Visibility is most fragile in workplaces and other power-asymmetric settings. Participants described being put on the spot when the avatar surfaced at the wrong moment (e.g., a manager asking “Everything okay?” after an orange face). Here, a playful cue becomes a compelled disclosure. This indicates that any socially legible affective display needs an easy way to soften, hide, or hold the cue in contexts where interpretation is high-stakes.

Worry spirals under persistent “red.” Several accounts show a different failure mode: prolonged “high stress” that does not match felt experience can trigger rumination, searching, and anxiety (e.g., Googling HRV interpretations for hours). This is a concrete form of breakdown: the cue is legible (red) but not interpretable (why red), reducing agency [29, 31]. Safeguards must therefore target persisting red specifically (e.g., explain-on-demand, reassurance without diagnosis, and off-ramps that redirect toward low-stakes actions rather than more monitoring).

Moralization through comparison. Competition often stayed friendly precisely because stakes were small (coffee, boba, chores) and groups maintained a norm that the cue was contestable. But the same comparison mechanic can moralize stress (“losing”), especially if repeated reds are read as personal failure or if group norms shift toward judgment. Participants opted out of comparisons or reframed challenges as cooperative rituals when pressure rose. Designers should avoid defaults that reward “being calm” as a stable trait and instead support cooperative framing (time-boxed, opt-in, solidarity cues).

Latency confusion. A subtler failure mode arises when participants cannot tell whether a mismatch reflects latency or error. Treating a delayed signal as a current verdict (or vice versa) leaks the interpretive flexibility that ordinarily sustains play, because the cue is being read against the wrong moment. Designs that surface a coarse sense of *when* a state was inferred (rather than only *what* state is shown) can preserve contestability without forcing users to manage timing themselves.

These boundary conditions strengthen the central claim: in this case, ambiguity helped sustain play only when representation and sharing mechanics kept interpretive stakes low and consent intact.

5.4 Scope of the Contribution

Our contribution is bounded. We do not claim that all biosignal wearables function as ambiguity-sustained ludic systems, nor that ambiguity is always a resource. The case we examine combines a specific representational form—an avatar-based, glanceable, watch-face placement with qualitative states—with specific sharing affordances (wrist showing, screenshot circulation) and a specific inference domain (stress) where mismatches are common. Each of these features carries weight. A numeric dashboard buried in a phone app would likely produce different practices, because visibility and circulation would not be doing the same work; a precise, well-validated cue would shift effort away from interpretation and likely toward score-style comparison.

This also means the contribution sits beside, rather than against, prior work on ambiguity, gamification, and social translucence. More precisely, our findings extend the ambiguity-as-resource perspective by showing that ambiguity operates not only as an individual interpretive prompt but also as a social mechanism through which groups co-construct meaning. They also complicate accounts of social translucence and interpersonal awareness by showing that affective ambiguity requires stronger audience controls and interpretive scaffolding: the same visibility that supports empathy in trusted ties can produce misreading or compelled explanation in power-asymmetric contexts. From an ambiguity-as-resource perspective [2, 14], our data show that ambiguity becomes livable through three practices: domestication through humor, collective calibration, and micro-actions that treat the cue as a prompt rather than a verdict. From a social translucence perspective [11], the avatar offers a wrist-worn version of the awareness–accountability dynamic, where translucence is supportive (empathy, coordination) when ambiguity is held in trusted ties and risky (misreading, compelled explanation) when it leaks into power-asymmetric audiences. Designing translucence for affective ambiguity therefore requires stronger audience controls and interpretive scaffolding than systems built on more stable signals.

5.5 Limitations and Reflexivity

Our study has limitations that shape how the claims should be read.

Instrumental case and design specificity. We focus on a single consumer app (*StressWatch*) as an instrumental case. The avatar-based, glanceable watch-face representation is central to the play practices we observed; results may not transfer to stress trackers that rely primarily on numeric dashboards, hidden menus, or phone-only views. Our contribution is therefore not a general theory of all stress wearables, but an account of how a particular representational form (a socially legible avatar) can enable ambiguity-sustained play.

Perceived mismatch rather than clinical ground truth. We analyze *perceived* mismatch between inferred stress and felt experience, not physiological or clinical validity. We did not audit the app's algorithm, sensor pipeline, or ground truth stress. Accordingly, our findings should be interpreted as evidence about interpretation, appropriation, and social meaning-making around a contested cue—not as evidence about whether the system accurately detects stress.

Sampling bias and visibility effects. Our interview recruitment (social media and snowball sampling) likely over-represents engaged users and may under-represent critics or those who abandoned the app early, especially in interviews. Moreover, reviews and public posts capture moments people chose to share, which may amplify social and performative practices relative to mundane private use.

Cross-context comparison is intertwined with platform ecology. Differences between China and North America in our corpus are inseparable from platform affordances and norms (public posting/commenting on RED versus more private messaging-based circulation in North America). We therefore do not claim cultural generalizations; instead, we treat these settings as two sharing ecologies that help surface how audience and visibility shape appropriation.

Counterfactual inference about accuracy and aesthetic. Our argument that contestability does work beyond the playful aesthetic rests on patterns within our data—which forms of play turned on *mismatch* rather than on the cute character (Narrative and Reflective Play in particular)—rather than on direct comparison with a more accurate system or with a less playful design. We therefore make this claim as inference, not observation. A controlled comparison—an avatar-based stress display

with substantially better calibration, or a non-playful display rendering the same state—would let the field test whether the dynamics we describe scale or shift.

Future directions. Future work could intentionally recruit dissatisfied and lapsed users, pair interviews with longitudinal in-situ diaries to capture everyday private interactions alongside share-worthy moments, and extend to other apps and representational styles to test how play varies with interface mechanics. Where appropriate, optional ground-truthing or sensor-quality audits could help relate perceived mismatch to signal quality and context, without collapsing the analysis into a validity evaluation.

6 Implications for Design

Our findings suggest that avatar-based, qualitative, and socially visible stress wearables like *Stress-Watch* can function less like measurement dashboards and more like *playable social artifacts*: people show them, compare them, narrate them, care for them, and probe them through small experiments. Designing for this possibility requires treating ambiguity, visibility, and social circulation as *core mechanics* rather than as side effects. Below we translate our themes into game-designable interaction loops paired with HDI-oriented safeguards.

6.1 Design Contestable Cues, Not Verdicts

When stress inference is inherently uncertain, interfaces should avoid “false precision” that invites diagnostic interpretation. Instead, represent stress as a *contestable cue*: a lightweight state that explicitly leaves room for interpretation [7, 14]. To keep ambiguity playable rather than anxiety-inducing, pair expressive states with *legible uncertainty cues* and low-friction access to “what this might reflect” (e.g., recent movement, caffeine, sleep, or signal quality) without claiming ground truth. A simple pattern is *Explain-on-demand*: a tap reveals a small, plain-language panel that separates “physiological arousal” from “psychological stress,” shows confidence/quality indicators, and offers non-alarming interpretations. This supports Narrative and Reflective Play (folk theories and micro-experiments) while reducing worry spirals when a state persists.

6.2 Treat Visibility and Consent as Game Rules

Because the avatar is glanceable and socially legible, visibility becomes a mechanic that can enable playful coordination but also cause unwanted disclosure. Designers should therefore treat *audience control* as a first-class rule system: users should be able to decide *when*, *to whom*, and *at what granularity* states are visible. Concrete patterns include: (1) a one-tap *privacy toggle* (hide/soften/hold) for meetings or public settings; (2) *audience modes* (private, close-ties, group) that change both what is shown and how often it updates; and (3) *contextual defaults* that avoid surfacing salient “red” cues in settings where interpretation is likely to be high-stakes (e.g., workplaces). These controls preserve Performative and Competitive/Cooperative Play while reducing social risk.

6.3 Support Emergent Play Loops Instead of Imposed Gamification

Participants did not primarily engage through points or prescribed challenges; they created their own stakes and rituals around showing, comparing, teasing, and caring. Rather than layering generic gamification on top, designers can support *emergent play loops* by providing “playable materials”: (i) *Showability* (quick reveal gestures, shareable snapshots), (ii) *remixable framing* (stickers/captions, state history collages, lightweight annotations), and (iii) *tunable stakes* (optional, time-boxed group check-ins or cooperative cool-down rituals). The goal is not to enforce a correct interpretation of stress, but to make it easy for people to create low-stakes social practices that fit their relationships and routines.

6.4 Design for Companionship Without Creating a Needy Companion

Many users related to the avatar as a pet-like character, which helped create emotional distance (“it is being dramatic”) and made coping actions feel gentle. However, persistent negative states also risk making the companion feel accusatory or “needy,” increasing checking and burden. Designs can preserve the benefits of companionship while reducing pressure by adding *downshifting mechanics*: the character can “rest” or become neutral after prolonged “red” periods; users can “pause” the companion without penalty; and the interface can encourage supportive micro-actions that do not require repeated monitoring (e.g., a single “take a breath” interaction that does not immediately re-score). Companion designs should aim for *soft attachment*—warmth and play without demanding vigilance.

6.5 Add Anti-Spiral Safeguards for Ambiguity Burden

Ambiguity can invite playful inquiry, but it can also externalize interpretive labor and amplify anxiety. Systems should include safeguards that detect and mitigate “spiral” conditions, such as prolonged high-stress states, repeated checking, or sudden state flips without context. Useful patterns include: (1) *Reassurance without diagnosis* (“this is an estimate and can be affected by movement, caffeine, or sensor fit”), (2) *redirects to low-stakes actions* (walk, water, breathing) rather than additional measurement, and (3) *off-ramps* (easy disengagement and reminders that opting out is acceptable). These safeguards align with HDI principles of legibility and agency while maintaining space for playful interpretation.

6.6 Evaluate for Playable Experience and Social Care, Not Only Accuracy

Finally, for avatar-based biosignal displays with similar conditions, evaluation may need to extend beyond accuracy and adherence metrics. Designers and researchers can assess whether a system supports: (i) *interpretive agency* (can users contest and reframe cues?), (ii) *consentful social circulation* (can users manage audiences and disclosure?), and (iii) *supportive play* (do interactions enable humor, care, and gentle coordination without moralizing stress?). For systems with similar representational and social affordances, the goal is not merely to detect stress, but to make stress *socially and personally workable* in everyday life.

7 Conclusion

Wearable stress tracking is often framed as a measurement problem: can a device infer an internal state, and will people follow its guidance? This paper shows a different possibility. Studying *Stress-Watch*, a smartwatch app that renders stress through a glanceable avatar, we find that contestable stress feedback can become a *playable* social resource in this instrumental case. Across App Store reviews, public posts on RED (XiaoHongShu), and 18 interviews, users repeatedly treated the avatar as something to *perform* (showing and staging states), *narrate* (jokes, folk theories, and worldbuilding), *play with together* (low-stakes challenges and solidarity rituals), *care for* (pet-like companionship), and *probe* (micro-experiments and pattern-seeking). These play forms emerged because the cue was socially legible, easy to circulate, and open to interpretation, allowing ambiguity to function as a resource for social play.

We argue that, in this instrumental case, the avatar functions as an *ambiguity-sustained ludic system*: its meanings are produced through interaction, negotiation, and shared practices rather than through precision alone. This suggests directions for studying biosignal displays that share its representational features: qualitative states, expressive characters, and glanceable watch-face placement. The mechanism we identify is the playable gap between system output and lived experience, made actionable through four mechanics—visibility, contestability, circulation, persistence—and enacted

in five empirical play forms. At the same time, ambiguity and visibility introduce risks—including worry spirals, social pressure, and unwanted disclosure—that become especially salient when cues are read as verdicts. We therefore highlight design principles that treat contestability and consent as core mechanics: legible uncertainty, explain-on-demand support, audience and visibility controls, and anti-spiral safeguards that preserve playful flexibility while supporting care.

By reframing this avatar-based stress wearable through a lens of play, our work contributes to broader CHI PLAY discussions of where play lives: not only in games, but in the everyday negotiations people build around ambiguous feedback, social relationships, and the wearable devices on their wrists.

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A Interview Protocol (Representative Questions)

The semi-structured protocol below was used as a guide rather than a script; follow-up probes were tailored to each participant's responses. The full protocol is available from the authors on request.

Background and routine use. Could you tell me how you first started using *StressWatch*? In a typical day, when do you find yourself looking at the avatar, and why? Are there moments when you intentionally check it, and moments when you happen to notice it?

Memorable incidents. Can you describe a time when the avatar's reading really stood out to you—either because it surprised you, made you laugh, worried you, or felt particularly accurate? What happened next?

Perceived mismatch and trust calibration. Have there been times when what the avatar showed didn't match how you actually felt? Could you walk me through one of those moments? What did you make of it? Did it change how you think about the app over time?

Responses to prompts/alerts. When the avatar changes state or the app prompts you, what do you usually do (if anything)? Are there situations where you ignore it or actively dismiss it? Why?

Sharing practices. Do you ever show the avatar to family, partners, friends, or colleagues, either in person or through a screenshot? Could you describe a recent example? What did the other person do or say?

Group and ritual practices. Are there any small routines or running jokes you (or people around you) have built around the avatar—comparisons, check-ins, challenges, or anything else? How did those start?

Impacts on wellbeing, relationships, and work/study. Looking back over your time with the app, in what ways has it affected how you feel, how you relate to people around you, or how you handle work or study? Are there contexts where you'd rather not have it visible (e.g., meetings, public settings)?

Closing. Is there anything we haven't talked about that you think is important for understanding how you use the avatar? If you were redesigning this kind of app, what would you change or keep?

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